MARK SCHEME · TEACHER COPY

12BR · Topic 17: Security

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Total marks: 31

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 31 marks

Question 1 · Oct/Nov 2025 Paper 31, original Q9 (a),(b) [4 marks]

Question	Answer	Marks
1(a)	**One** mark for each mark point (**Max 2**) MP1 Ensure security/privacy when using the internet MP2 Data encryption MP3 Identification / authentication of client and server	2
1(b)	**One** mark for each mark point (**Max 2**) MP1 When transmitting authentication data e.g. passwords, session cookies MP2 When transmitting data that must be protected from modification on its way to or from a server e.g. user input, or results from the server MP3 When transmitting data classified as non-public.	2

Question 2 · May/June 2026 Paper 33, original Q8 (a),(b) [6 marks]

Question	Answer	Marks	Guidance
2(a)	**One** mark per mark point for the purpose (**Max 2**) MP1 To provide a more secure method of cryptography/encryption than current methods // To provide a virtually unbreakable method of encryption // To detect eavesdropping. MP2 … using the laws of quantum physics rather than mathematical formulae to generate keys. MP3 To provide encryption that can be guaranteed for a longer period of time than current methods.	2	

Total: 31 marks

2(b)	**One** mark per benefit (**Max 2**) MP1 Better security for internet communication MP2 Can detect eavesdropping, enabling the receiver to request another key MP3 Virtually unhackable due to rapidly changing quantum states MP4 Keys are non-duplicable // No-cloning theorem **One** mark per drawback (**Max 2**) MP5 Currently can only be used over relatively short distances MP6 Very expensive to implement initially due to costs of installation / hardware / implementation // Requires a dedicated line and specialist hardware MP7 The technology to fully implement quantum cryptography is not currently/readily available MP8 Photon polarization might be impacted in the travelling medium // It is possible for the polarisation of the light to be altered while travelling down fibre optic cables MP9 Due to the inherent security system generated by quantum cryptography, terrorists and other criminals can use the technology to hide their activities from government law enforcement	**4**	

2	W⇨N2⇨N5	E	4+7+11=22	0	22

Question 3 · May/June 2026 Paper 32, original Q8 (a),(b) [7 marks]

Question	Answer	Marks	Guidance
3(a)	**One** mark per mark point (**Max 3**) MP1 Symmetric key encryption uses only one key but asymmetric encryption uses two keys // Asymmetric public and private keys are a mathematically linked pair, whereas symmetric uses the same key. MP2 Asymmetric key encryption uses one key to encrypt and the other key to decrypt, whereas symmetric key encryption uses the same key for both. MP3 Symmetric key encryption suffers from the key distribution problem, which the matched pair of keys for asymmetric has resolved. MP4 Asymmetric keys must be much larger/longer to achieve the same level of security as symmetric key encryption. MP5 Asymmetric key encryption allows verification of the sender, which symmetric key encryption does not.	**3**	Candidates are asked to describe the difference, so both key types need to be included for each mark, where applicable.

3(b)	**One** mark per mark point (**Max 4**) MP1 The organisation generates a matched pair of encryption keys MP2 The organisation makes its public key available to the group of people / via its website MP3 The organisation encrypts its message using its private key MP4 … and sends it to the group of people MP5 Anyone receiving the message can decrypt it using the organisation's public key MP6 Therefore, verifying the **sender** // Providing non-repudiation.	**4**	The correct sequence of steps that is important.

Question 4 · Oct/Nov 2025 Paper 32, original Q10 [4 marks]

Question	Answer	Marks
4	**One** mark for identifying a protocol and **one** mark for stating its purpose MP1 Handshake protocol MP2 To establish a secure and reliable connection between two devices, systems or networks // Permits the web server and client to authenticate each other to make use of encryption algorithms MP3 Record protocol MP4 Provides a secure and reliable way to send and receive data over a network // To exchange records between the client and server // Responsible for securing application data end ensuring its integrity and authenticity during transmission // Encrypts and authenticates data exchanged between a client and a server // Deals with the format for data transmission.	**4**

Question 5 · May/June 2026 Paper 31, original Q8 (a),(b) [6 marks]

Question	Answer	Marks	Guidance
5(a)(i)	Text **encrypted** from plain text using an encryption algorithm // The result of putting plain text through an **encryption algorithm**.	**1**	

5(a)(ii)	A cryptographic/encryption key that can be obtained/used by anyone to send encrypted messages. // A key that is part of a matched pair // An encryption key known to all users // A decryption key obtained/used by anyone to decrypt sender's digital signature	**1**	
5(b)	**One** mark per mark point (**Max 4**) MP1 The organisation generates a matched pair of encryption keys MP2 The organisation makes its public key available to the individual MP3 The individual uses the public key to encrypt their message MP4 … and sends this message to the organisation MP5 The organisation decrypts the message using their private key **Alternative answer** **One** mark per mark point (**Max 4**) MP6 The individual generates a matched pair of encryption keys MP7 The individual sends their public key to the organisation MP8 The individual uses their private key to encrypt their message MP9 … and sends this message to the organisation MP10 The organisation decrypts the message using the individual's public key	**4**	The correct sequence of steps that is important.

Question 6 · Oct/Nov 2025 Paper 33, original Q7 [4 marks]

Question	Answer	Marks
6	**One** mark for each correct marking point (**Max 4**) MP1 An SSL/TLS connection is initiated by an application/client. MP2 Every new session begins with a handshake as defined by the SSL/TLS protocols. MP3 The client requests the digital certificate from the server // The server sends the digital certificate to the client. MP4 The client verifies the server's digital certificate MP5 … and obtains the server's public key. MP6 The encryption algorithms are agreed. // The symmetric session keys are generated/defined. MP7 A secure session is established between client and server.	**4**

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